The use of artificial intelligence in engineering tools

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Abstract. The following work aims to illustrate that the artificial intelligence has penetrated not only into major fields like manufacture, agriculture, entertainment, but also well coexist with the engineering software. Engineering fields are no exception when integrating the technology into its processes. The embedment of artificial intelligence in the processes allows increase in productivity and possibilities for innovative approaches and for new breakthroughs. This work will look into the artificial intelligence and its use in engineering fields. Focusing the view on the implementation of the technology in the analytical tools. The latest increase of the capabilities of such software is directly related to the improvement in the field of algorithm. Henceforth, this work will dive into the new features and capabilities this technology presents. The work will cover the areas in engineering tools which are enhanced with artificial intelligence. The following areas will be covered: generative design, multi-objective optimisation and personalisation of the using tool to meet the consumers' needs in the most responsible ways.

1 Introduction

Fast changing world introduces new innovative approaches to be implemented in various ways. Unimaginable automatization is achievable by applying these novelties. It is believed that in the near future, almost every process will be dominated by mechatronics. Less human interaction is the outcome of such technological progress. There is no clear vision on how it will effect the humans lives. However, technologies are improving and similarly easing one's lives, if looking through one perspective. Any technological progress is followed by global changes depending on the technology and its adaptation. It brings not just a positive side, but there is another if looked into thoroughly. This work will cover the use of artificial intelligence in engendering field. With the narrowed focus on analytical tools, which are used to simulate the given structure with boundary conditions on different parameters.

Why is engineering field is important and hence its improvement and refining. One can refer to its involvement in our daily life and the easiness it brings with its each iterative approach. Every new detail provided by this field enriches our existence. It elevates our capabilities in various areas and provides a helpful aid in different circumstances. Therefore,

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any further refinement is welcomed and easily adopted for better practices. In this work, artificial intelligence will be covered as an area that will further benefit to this field.

Artificial intelligence is one of the most significant technology that emerged from algorithmic improvement. It is a phenomenon that will defy our future. Its influence is already felt by different industries, fields, and areas. Humans are greatly influenced too. It penetrated unnoticeable every possible process that leaves most of the people unaware of its presence. In a short time, the technology achieved more than other technologies could achieve. Although, it must be said that the technology is a combination of different technology, methods, and approaches. Alone, it is just an algorithm written in a more complex way. Combined with others, it can achieve the best results [1, 2, 3].

2 AI and how it is moved to engineering

The use of artificial intelligence in engineering field is an inevitable outcome. Right now, it is well embedded in most of the engineering tools (CAD). Due to the possibilities of the technology, it was adopted in several areas that will be covered in this work. Why this technology is greatly adaptable in various fields and areas. The first thing is related to its flexibility. The technology has no constrain, meaning that it can be moulded into any shape. Of course, it has no physical shape if observing algorithmic consistence. Although, it has limitation in a physical shape, as development of those two are in different levels. Artificial intelligence is a combination of algorithms that are built for a certain area of applications. It is complex mathematical models that are built to mimic human capabilities. In some of the task they are far ahead humans. Repetitive task, information analysis, pattern recognition, generation of data and many more. There is another side of the technology in which it lacks abilities to imitate human behaviour. It is well known that the technology is in its first stage of improvement. In the near future, it will get more intelligent and surpass human in many aspects. However, even with its first stage it is capable of refining the given field or area to an unimaginable way.

As it was mentioned before, the technology was adopted due to its capabilities to automate processes with less human environment. The nature of humans is to ease physical or mental burdens. The use of artificial intelligence provides less human interaction in different processes while increasing the quality of the task. Similarly, engineering fields adopted this technology to automate and to increase the quality of the end results. The following paragraph will be covering the use of artificial intelligence in engineering tools. Which aspect were contributed in the transformation and which received more advantages from the use [4].

3 AI in engendering tools

In this paragraph, artificial intelligence use in engineering tools will be covered. These tools are used to build, assembly and analysis. There are other steps that can be done by these tools. However, this work will only focus on these three stages. There are many tools that can perform abovementioned steps. For instance, SolidWorks, Inventor, Fusion 360, Ansys and so on. These tools are greatly appreciated in engineering fields and are utilized by many researchers. Embedment of artificial intelligence become gradually. Some of the companies does not do the accent on the use of artificial intelligence in their products. However, other build around it the whole marketing strategies. Most of the time, the technology is used for some optimization of small processes. However, the main use of the technology in this field is structure optimization. Structure optimization is a time-consuming process in which different structural parameters achieved. Due to the use of artificial intelligence, this process can be eased in many different aspects, including the providence of various shapes of one

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structure. Generative design and multi-objective optimization methods can be outlined when using artificial intelligence in engineering tools [5, 6].

3.1 Generative design

As we all know, artificial intelligence is mostly used to generate data. It can generate text, images, videos, etc. The technology can also be used to generate three-dimensional models. Hence, the technology is embedded in the iterative process where the necessary shapes are generated. In numerical analysis, it is used through different approaches to find the optimal structure appearance. In the past, shape optimization or sensitive analysis was done manually. Each iteration required manual change of parameters, which would lead to the completely wrong path. Researchers would spend days and weeks to find optimal solutions. However, in the use of algorithms, specifically made for the structure analysis, one could easily achieve desired outcomes. Generative design is capable of providing unconventional, creative and highly efficient design. Artificial intelligence accelerates and automates the iterative steps to explore a wide range of shapes in respect to the provided constrains, boundary conditions and other objectives. How does it work? Built part or assembled structure is constrained, and boundary conditions are added. After simulation, one will be provided with a clear visual representation of different parameters. Hence, the structure can be improved by eliminating unnecessary material usage, increasing the robustness in the areas with most distraction, providing various shapes that suit the aim, etc. This technique can also predict the weak spots and possible regions within the structure that will fail over the time. It can suggest how to dissipate the pressure and forces correctly, so the structure will withstand the given loads [7, 8].

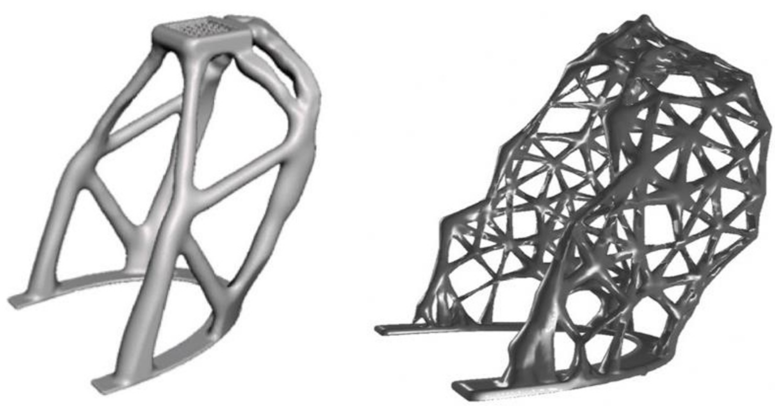

Fig.1. Generated design with Generative design [9]

3.2 Multi-objective optimisation

Another technique which uses artificial intelligence to enhance its capabilities is multi-objective optimization. This one is similar to generative design in its approaches. However, there are some dissimilarities if studied thoroughly. Multi-objective optimization uses artificial intelligence to enhance the design and decision-making process. Multi-objective optimization is used when there is no one clear solution, but different opposing goals must be met. The technique must identify the optimal trade-offs between conflicting objectives. Most of the time it is related to the following objectives: quality, cost, efficiency, environment impact and so on. Therefore, by including all the conflicting objective it must provide a set of solutions, which are known as Pareto Front. This is a fundamental concept in multi-objective optimization. It states that no one objective can be improved without degrading at least one of the other objectives. Henceforth, the multi-objective optimization utilizes artificial intelligence to analysis different objectives to reach the most optimal solution with all the necessary step taken automatically [10].

3.3 Personalised interaction

The second thing that provides artificial intelligence is personalized experience. It will provide the researcher the most personalized settings when using engineering tools. It is well known that most of the time a certain researcher will perform the same tasks over and over. Repetitive task takes most of the time in the whole process. Therefore, the technology embedment comes in handy when using engineering tools for different analysis. Algorithms will analyze the behaviors of potential users and then provide the most suited experience. It will perform some steps automatically after a while. Some of the tools will dominate over others as their preference. Therefore, the use of artificial intelligence in analytical tools will level up the longevity of the process to ensure the best possible personalized experience [11, 12].

4 Future perspective

It is always hard to foresee the tendency of a certain technology. It is almost impossible to predict the path of a technology that is applicable in a wide range of fields and areas and which is expanding in terms of application and improvement. However, analyzing its past and present tendency and its influence, some suggestion on it future improvement can be indicated. This paragraph will look at the future perspective of the technology through engineering tools perspective. The most obvious changes will be in automatization of whole processes in analytical tools. The technology will provide the ease in all stages by minimizing the researcher involvement and maximizing the quality of end results. New capabilities will be brought in that allows more thorough analysis of a given structure. It is also possible that the interaction between tools and researchers will be improved. Meaning that researchers could use text or prompts to perform the analyzing. This will eliminate the need to search for the necessary information on how to perform a certain analysis and the need for programming. To guess the exact possible outcome is impossible when dealing with complex technology, but some of the paths can be outlined [13, 14, 15].

5 Conclusion

The following work was done to illustrate the use of artificial intelligence in engineering fields. The focus was done on the tools that are used for numerical analysis in engineering fields. Artificial intelligence is well known for its multi-functionality and its capabilities to automate various field to the next level. In engineering fields, it can be used for many areas, but the main application is the structure optimization. This technology can automate stems involved in numerical analysis to provide the most unconventional and high-quality shapes. The work went through two main areas that utilized the technology. The two areas are generative design and multi-objective optimization. The last paragraph of this work was devoted on the future perspective of the technology in the aforementioned two areas.

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